

CHAPTER 3

ICT SKILLS - II

INFORMATION TECHNOLOGY (402)
ARTIFICIAL INTELLIGENCE (417)

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ICT SKILLS

Introduction to operating system :-

Operating System (OS) is the system software that performs all the basic tasks of a computer system. It works like an interface between the user and the computer hardware resources. It actually controls the execution of all the tasks that are performed in a computer.

Operating System

Some examples of operating systems are Unix, MS-DOS, Windows 98/2000/XP etc. It enables a user to utilise hardware resources very efficiently. So, operating system makes it easy for a user to use computer system.

"An operating system is a collection of programs that controls and coordinates all the tasks and functions in a computer system. It mediates between application programs and hardware components."

Operating system is the main part of the system software of any computer. It mainly provides an environment to run the software and serves services to the computer hardware.

Basically, operating system has two objectives :

- **Managing the computer's hardware**

And

- **Providing the interface.**

Functions of Operating System

The main functions of an operating system are as follows :

- Process Management :- A process is an application or a program in running state. Process management is the important part of an operating system which enables the activities of planning, monitoring and performance of a CPU.
- Memory Management :- As a memory manager, the operating system handles the allocation and deallocation of memory space as required by various programs. Managing the primary memory, sharing and minimising memory access time are the basic goals of the memory management.

Functions of Operating System

- File Management :- The operating system is responsible for creation and deletion of files and directories, It also takes care of other file related activities such as organising, storing, retrieving, naming and protecting the files.
- Input/Output Management :- The input/output management module of an operating system coordinates and assigns different input and output devices. It controls all input/output devices, keeps track of input/output requests, issues commands to these devices.

Functions of Operating System

- Security Management :- The operating system protects system resources and information against destruction and unauthorised use , it keeps different programs and data in such a manner that they do not interface with each other.

Operating Systems for Desktop and Laptop

Some of the commonly use operating systems for desktops and laptop are:

- **UNIX** : It was **developed in the late 1960s as a multi-user, multitasking operating system that supports both command-line and graphical user interface (GUI)**. It was written in the C programming language, which enabled its portability across various hardware platforms.
- **UNIX is known for its security, stability, and multitasking capabilities, making it popular in universities, servers, and research systems.**

Operating Systems for Desktop and Laptop

- **LINUX: Open-source operating system inspired by UNIX.** Available for free through various distributions (e.g., Ubuntu, Fedora). **It is known for its high security, stability, and resistance to viruses.**
- The system supports a wide range of hardware and software, making it ideal for programming, networking, and server use.
- **Mac OS (Apple Macintosh):** It was developed by Apple Inc. and introduced on March 24, 2001. **This operating system is based on UNIX, which provides it with a stable and user-friendly graphical interface, making it less prone to crashes and bugs.**
- **However, a notable limitation of macOS is that it only works on devices manufactured by Apple.**

Operating Systems for Desktop and Laptop

- MS-Windows (Microsoft Windows): **Windows is a Graphical User Interface (GUI) based operating system by Microsoft**, first released as an add-on to MS-DOS in 1985.
- It has since evolved through **multiple versions, becoming the most widely used OS for personal computers.**

Versions of Windows

Windows NT (New Technology)

- Released: July 27, 1993
- Processor-independent, multi-user, and multi-processing system.
- Used mostly in business and networked environments.

Windows 95

- Released: August 1995
- First major consumer-oriented Windows OS.
- Introduced the Start menu and taskbar.
- Merged Windows with DOS into a single OS.

Windows 98

- Released: 1998
- Came in two versions; second edition was more stable.
- Improved hardware support and USB functionality.

Windows 2000

- Released: February 2000
- Built for both home and business use.
- Improved file system (NTFS) and better networking.

Windows ME (Millennium Edition)

- Released: September 2000
- Successor to Windows 98.
- Included multimedia improvements but was buggy and unstable.

Versions of Windows

Windows XP (eXPerience)

- Released: October 25, 2001
- Very popular version with Home and Professional editions.
- Combined Windows 2000's stability with a user-friendly interface.
- Longest-running Windows version in active use.

Windows Vista

- Focused on enhanced security, appearance, and media tools.
- Introduced User Account Control (UAC).
- Criticized for being slow on older machines.

Windows 7

- Improved performance over Vista.
- Enhanced interface and better compatibility.
- Widely adopted and praised for stability

Windows 8

- Introduced in 2012 with touchscreen support.
- Featured a new Metro UI replacing the Start menu.
- Mixed reviews due to drastic interface changes.

Windows 10

- Released in 2015.
- Unified platform across PC, tablet, and mobile.
- Continuous updates and improved user interface.

Operating Systems for Desktop and Laptop

- Solaris: It was developed by Sun Microsystems in 1993 and later acquired by Oracle, is a **UNIX-based operating system celebrated for its exceptional scalability, performance, and security features.**
- It has become a **popular choice for enterprise servers and network environments, where these attributes are crucial for managing complex and demanding tasks.**

Operating Systems for Desktop and Laptop

- MS-DOS, or Microsoft Disk Operating System, It was developed by Microsoft specifically for IBM PCs in the early 1980s. As a text-based, single-user operating system, it served as the foundation for the initial versions of Windows, preceding Windows 95.
- Over time, however, MS-DOS was phased out in favor of more user-friendly graphical user interface (GUI)-based systems.

Operating Systems for Mobile Phones and Tablets

A Mobile Operating System (OS) controls smartphones, tablets, and mobile devices. It manages wireless communication, app usage, and multimedia support.

Operating Systems on different devices:

1. Android

- **Developer: Google (2008)**
- **Type: Linux-based, open-source**
- **Devices: Touchscreen phones, tablets**
- **Popularity: Most widely used mobile OS**

Operating Systems for Mobile Phones and Tablets

2. iOS

- **Developer: Apple Inc. (2007)**
- **Devices: iPhone, iPad, iPod Touch, Apple TV**
- **Features: Closed-source, optimized for Apple devices**

3. BlackBerry OS

- **Developer: BlackBerry Ltd. (1999)**
- **Known for: High security, supports WAP 1.2**
- **Usage: Mainly in enterprise-level phones**

Operating Systems for Mobile Phones and Tablets

4. Symbian

- **Developer: Symbian Ltd. (1997)**
- **Type: Open-source**
- **Features: Good PIM (Personal Info Mgmt), supports various manufacturers (Nokia, Sony, etc.)**

User Interface(UI)

Character User Interface (CUI)

- A Character User Interface (CUI) is a text-centric interface that requires users to input commands using a keyboard.
- It is characterized by its reliance on textual input for interaction rather than graphical elements.
- This type of interface is exemplified by systems like MS-DOS and various Command Line Interfaces (CLI), where users can directly engage with the operating system through written commands.

User Interface(UI)

Graphical User Interface(GUI)

- The Graphical User Interface, commonly known as **GUI**, utilizes icons, windows, and pointing devices to enhance user interaction.
- This **design approach** makes technology far more user-friendly compared to previous interfaces.
- The concept was initially developed by Xerox in the 1970s and has since become a fundamental component of various devices, including smartphones and media players.
- By providing an intuitive means of navigation, **GUIs** have revolutionized the way we interact with technology today.

QUESTIONS

Question : Mention any two functions of Operating System.

Question : When was Windows 10 Operating System released?

Question : Name any two mobile Operating System.

Question : What is the full form of GUI?

Windows 10

- **Windows 10, developed by Microsoft and released in 2015,** is known for its user-friendly interface that appeals to a wide range of users.
- **One of its notable features is Game DVR, which allows players to easily record their gameplay.**
- **Additionally, the interface of Windows 10 is designed to adapt seamlessly to the hardware it's installed on, ensuring a smooth and optimized user experience across various devices.**

THANK - YOU

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